

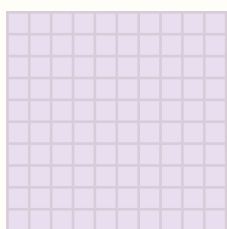


Understanding place value

Place value gives every digit its meaning

A digit's value depends on its position. In 346, the digit 3 represents three hundreds, 4 represents four tens and 6 represents six ones. This understanding helps pupils read, write, compare and calculate with confidence.

SEE THE VALUE IN 346



3 hundreds



4 tens



6 ones

346

$$3 \times 100 + 4 \times 10 + 6 \times 1$$

$$= 300 + 40 + 6$$

Flexible partitioning

Pupils work with numbers up to 1,000, learning to break them apart and recombine them. For example, 146 can be written as $100 + 40 + 6$, or as $130 + 16$. Recognising equivalent partitions supports efficient mental and written calculation.

$$146 = 100 + 40 + 6$$

$$146 = 130 + 16$$

Different partitions, but exactly the same total.

Why it matters

Without place value, numbers are only symbols. With it, pupils can compare quantities, estimate answers, notice errors and choose efficient methods. These skills underpin addition, subtraction, multiplication, division, money, measurement and data.

CORE IDEA

Place value shows how numbers are built and how they can be changed.



Using numbers to 1,000

Everyday numbers

Numbers up to 1,000 appear in prices, measurements and quantities. Familiar examples help pupils connect written numerals with real amounts and make sensible comparisons.

£875

HOLIDAY COST

25 from 900; 75 from 800.
It is closer to 900.

320

SPORTS DAY

32 tens, or 3 hundreds
and 2 tens.

250 ml

RECIPE

Exactly halfway between
200 and 300.

Counting in useful steps

Counting from 0 in multiples of 4, 8, 50 and 100 develops pattern recognition and supports later multiplication and division.

4s 0 • 4 • 8 • 12 • 16 • 20

8s 0 • 8 • 16 • 24 • 32 • 40

50s 0 • 50 • 100 • 150 • 200

100s 0 • 100 • 200 • 300 • 400

Finding 10 or 100 more and less

Pupils should change a number mentally without counting in ones. They identify the place-value column that changes and the digits that remain the same.

$$430 + 10 = 440$$

tens change

$$430 - 10 = 420$$

tens change

$$430 + 100 = 530$$

hundreds change

$$430 - 100 = 330$$

hundreds change

By the end of this topic, pupils should be able to:

- recognise each digit's value in a three-digit number
- read and write numbers to 1,000
- compare and order numbers to 1,000
- represent and estimate numbers in different ways
- count in multiples of 4, 8, 50 and 100
- solve practical problems using these ideas